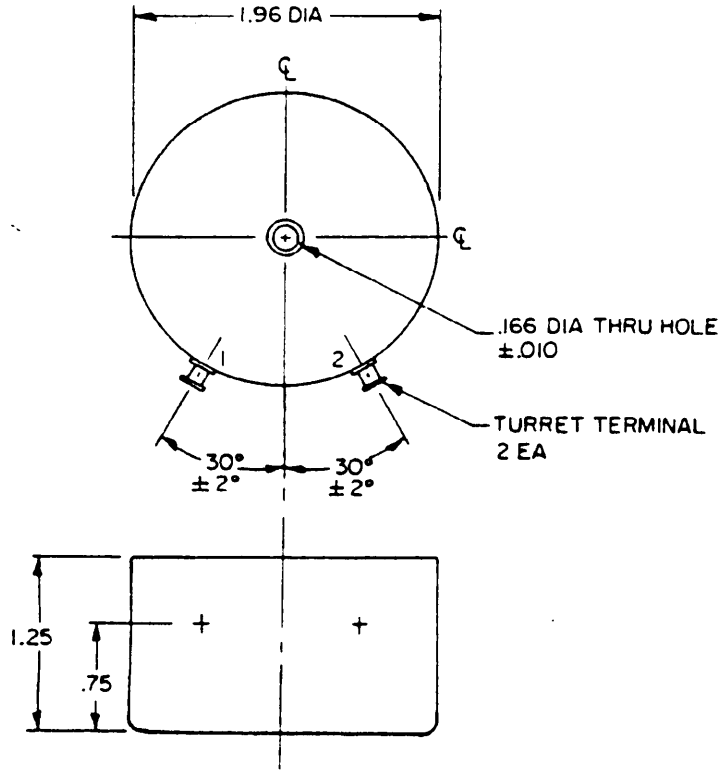
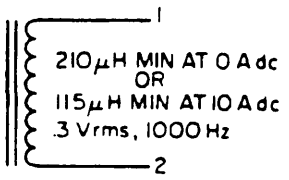
MILITARY SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS
(AUDIO, POWER AND HIGH-POWER PULSE),
INDUCTOR, POWER, TF5S04ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the inductor described herein shall consist of this document and the latest issue of Specification MIL-T-27.





Working voltage: 50 V Pk
Altitude: 62,000 Ft max
CIRCUIT DIAGRAM AND MARKING

INCHES	MM
.010	.25
.03	.8
.166	4.22
.75	19.0
1.25	31.8
1.96	49.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Marking shall be on the top of the case.
4. Tolerance on all dimensions shall be ±.03 unless otherwise specified.

FIGURE 1. Dimensions and configuration.

FSC 5950

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:01Z
Check the source to verify that this is the current version before use.

REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:
Inductance: 210 µH at 0 A dc or 115 µH at 10 A dc.
Voltage: 0.3 Vrms.
Frequency: 1,000 Hz.
DC resistance: 0.025 ohm, maximum.
Working voltage (peak): 50 volts.

Design and construction:
Dimensions and configuration: See figure 1.
Duty cycle: Continuous.
Case: Fiberglass filled with Diallylphthalate.
Terminals: Turret, hollow type, brass, to accept two wires 0.025 inch diameter (#22 AWG).
Height: .198 inch maximum.
Diameter: .110 ±.010 inch.
Wieght: 0.8 pound maximum.
Altitude: 62,000 feet maximum.
Operating temperature range: -55° to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds and condition E, 8 ounce-inches.

Shock: MIL-STD-810, method 516, procedure I, 15 G, half sine pulse.

Vibration (random): MIL-STD-202, method 214, test condition D, except the 6 dB/octave roll up shall intersect the 0.1 G^2/Hz point at 50 Hz.

Dielectric withstanding voltage:
At sea level: 100 volts rms.
At reduced barometric pressure: 100 volts rms.

Electrical characteristics:
Inductance: With 0.3 volts rms, 1,000 Hz and 0 amp dc applied to (1-2): 210 µH minimum or with 0.3 volts rms, 1,000 Hz and 10 amps dc applied to (1-2): 115 µH minimum.
Temperature rise: 42.5°C with 10 amps dc applied to (1-2) at an ambient temperature of 87.5°C.

Thermal shock: MIL-STD-202, method 107, test condition A-1.

Salt spray (corrosion): MIL-STD-202, method 101, test condition B.

Marking location: See figure 1.

Part number: M27/280-01.

Substitution data: This military transformer is substitutable for FSCM 72964, part number 19028. This in no way implies that the manufacturer's part is substitutable for the military item.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0561-18)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:01Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-27/280

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:01Z
Check the source to verify that this is the current version before use.